IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

CONTROL SYSTEMS

Wednesday, 28 April 10:00 am

Time allowed: hours

There are THREE questions on this paper.

Answer questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : S.A. Evangelou

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CONTROL SYSTEMS – 2020/2021

Recommendations for candidates:

The exam has to be carried out remotely following the official indications provided by the College. Moreover, the algebraic calculations have to be carried out using a standard calculator. The use of mathematical software such as Matlab and similar tools installed on personal computing devices is prohibited.

Information for candidates:

Remark for the Students: the information provided here is much more than needed to answer the Exam Paper Questions; that is, the presence of a specific formula does not mean that this formula is required to answer the questions.

- **Main Transforms**

$f(t)$	$F(s)$
$1(t)$	$\frac{1}{s}$
$t \cdot 1(t)$	$\frac{1}{s^2}$
$t^2 \cdot 1(t)$	$\frac{2}{s^3}$
$e^{\alpha t} \cdot 1(t)$	$\frac{1}{s-\alpha}$
$t \cdot e^{\alpha t} \cdot 1(t)$	$\frac{1}{(s-\alpha)^2}$
$\sin(\omega t) \cdot 1(t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t) \cdot 1(t)$	$\frac{s}{s^2+\omega^2}$
$t \cdot \sin(\omega t) \cdot 1(t)$	$\frac{2\omega s}{(s^2+\omega^2)^2}$
$t \cdot \cos(\omega t) \cdot 1(t)$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$
$e^{\sigma t} \cdot \sin(\omega t) \cdot 1(t)$	$\frac{\omega}{(s-\sigma)^2+\omega^2}$
$e^{\sigma t} \cdot \cos(\omega t) \cdot 1(t)$	$\frac{s-\sigma}{(s-\sigma)^2+\omega^2}$
$t \cdot e^{\sigma t} \cdot \sin(\omega t) \cdot 1(t)$	$\frac{2\omega(s-\sigma)}{\left[(s-\sigma)^2+\omega^2\right]^2}$
$t \cdot e^{\sigma t} \cdot \cos(\omega t) \cdot 1(t)$	$\frac{(s-\sigma)^2-\omega^2}{\left[(s-\sigma)^2+\omega^2\right]^2}$

Table 1 Time-functions and corresponding Laplace transforms

$f[k]$	$F(z)$
$1[k]$	$\frac{z}{z-1}$
$k \cdot 1[k]$	$\frac{z}{(z-1)^2}$
$k^2 \cdot 1[k]$	$\frac{z(z+1)}{(z-1)^3}$
$a^k \cdot 1[k]$	$\frac{z}{z-a}$
$k \cdot a^k \cdot 1[k]$	$\frac{az}{(z-a)^2}$
$\sin[\omega k] \cdot 1[k]$	$\frac{z \sin\omega}{z^2-2z\cos\omega+1}$
$\cos[\omega k] \cdot 1[k]$	$\frac{z^2-z\cos\omega}{z^2-2z\cos\omega+1}$
$a^k \cdot \sin[\omega k] \cdot 1[k]$	$\frac{az \sin\omega}{z^2-2az\cos\omega+a^2}$
$a^k \cdot \cos[\omega k] \cdot 1[k]$	$\frac{z^2-az\cos\omega}{z^2-2az\cos\omega+a^2}$

Table 2 Time-sequences and corresponding Z transforms

- **Miscellaneous**

– Maximum percentage overshoot: $\Delta\% = 100 \cdot e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}$

– Settling-time at 1%: $t_{s,0.01} \simeq \frac{4.6}{\xi\omega_n}$

1. a) Consider the block scheme shown in Figure 1.1:

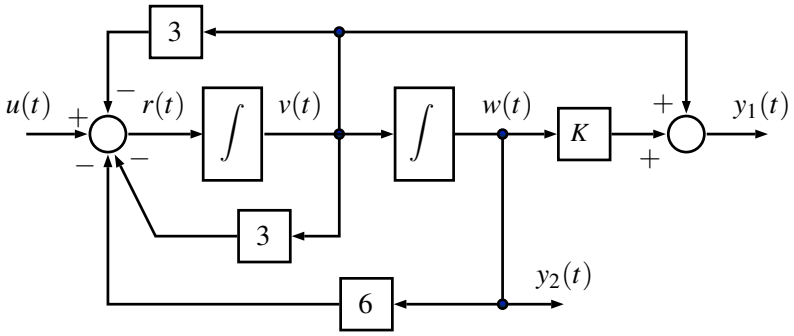


Figure 1.1 Block scheme for Question 1.

where $K \in \mathbb{R}$ is a given scalar, $u(t)$ denotes the input, and $y_1(t)$ and $y_2(t)$ denote the two outputs.

Determine a state-space description of the dynamic system shown in Figure 1.1 of the form:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

where $x \in \mathbb{R}^n$ is a suitably-defined state vector, $\dot{x}$ denotes its time-derivative, $u \in \mathbb{R}$ is the input function shown in Figure 1.1, $y = [y_1, y_2]^\top$ is the output vector whose components y_1 and y_2 are the output functions shown in Figure 1.1, and A, B, C, D are suitable matrices to be determined.

[12 Marks]

b) Establish the stability properties of the dynamic system shown in Figure 1.1 as a function of the scalar $K \in \mathbb{R}$.

[8 Marks]

c) Now, consider the linear time-invariant dynamic system described by the following state equations:

$$\begin{cases} \dot{x}_1(t) &= -x_1(t) + u(t) \\ \dot{x}_2(t) &= 0.5x_2(t) \\ \dot{x}_3(t) &= 0.2x_1(t) - 0.2x_3(t) \\ y(t) &= x_1(t) + x_2(t) \end{cases}$$

Setting

$$u(t) = 1(t-1); \quad x(0) = [0, 0, 0]^\top$$

determine the analytical expression of the output function $y(t), \forall t \in \mathbb{R}$, where $1(t)$ denotes the unit step function.

[20 Marks]

2. Consider the block scheme shown in Figure 2.1:

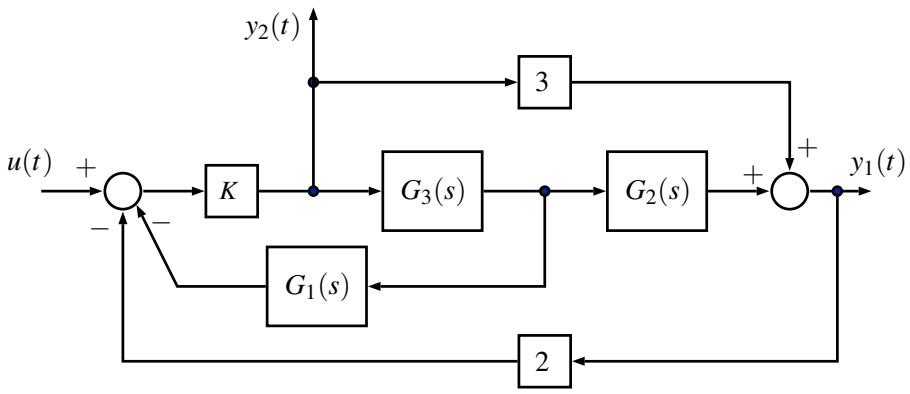


Figure 2.1 Block scheme for Question 2.

where, with some abuse of notation, we represent the inputs and outputs of each block in the time-domain instead of in the Laplace domain. Moreover, $K \in \mathbb{R}$ is a generic scalar, and $G_1(s)$, $G_2(s)$, and $G_3(s)$ denote generic transfer functions with no hidden common factors in their numerators and denominators.

a) Determine the transfer function $G_{u,y_2}(s)$ from the input $u(t)$ to the output $y_2(t)$ expressed in terms of the scalar K and the generic transfer functions $G_1(s)$, $G_2(s)$, and $G_3(s)$.

[8 Marks]

b) Suppose that $K = 0$. Establish whether the instability of either $G_1(s)$ or $G_2(s)$ or $G_3(s)$ implies the instability of the whole dynamic system described by the block scheme shown in Figure 2.1. Discuss your findings.

[6 Marks]

c) Now, suppose that $K \in \mathbb{R}, K \neq 0$. Analogously to Question 2-b), establish whether the instability of either $G_1(s)$ or $G_2(s)$ or $G_3(s)$ implies the instability of the whole dynamic system described by the block scheme shown in Figure 2.1. Discuss your findings.

[6 Marks]

3. Consider the block scheme shown in Figure 3.1:

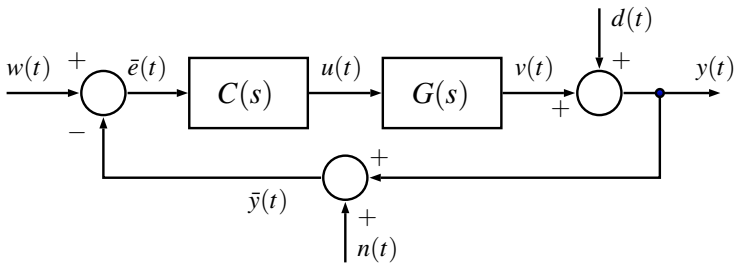


Figure 3.1 Block scheme for Question 3.

where, with some abuse of notation, we represent the inputs and outputs of each block in the time-domain instead of in the Laplace domain. Moreover

$$G(s) = 9 \cdot \frac{1 + 0.1\,s}{(1 + T\,s)}$$

The time-constant T is *unknown* and satisfies the constraint $T \in [0.5\,,\ 10]$.

a) Setting $d(t) = 0, \forall t$ and $n(t) = 0, \forall t$, design a controller $C(s)$ such that:

i) The closed-loop steady-state static error e_∞ satisfies:

$$w(t) = 1(t) \quad \Longrightarrow \quad \bar{e}_\infty = 0$$

ii) The settling time at 1% satisfies $t_{s,0.01} \leq 10$ sec

iii) The maximum percentage overshoot $\Delta\%$ satisfies $\Delta\% \leq 20\%$

iv) The phase margin φ_m satisfies $\varphi_m \geq 40^\circ$

If needed or deemed useful, use the logarithmic chart provided at the end of the Exam Paper.

[18 Marks]

b) Consider the controller $C(s)$ that you have obtained in answering Question 3-a) and set $w(t) = 0, \forall t$, $d(t) = 5 \cdot 1(t)$, and $n(t) = 0.1 \cdot \sin(\bar{\omega} t) \cdot 1(t)$.

Determine *approximately* the high-frequency range $[\omega^-, +\infty)$ such that

$$|y_\infty(t)| < 10^{-2}, \quad \forall \bar{\omega} \in [\omega^-, +\infty)$$

where $y_\infty(t)$ denotes the regime (steady-state) output function.

If needed or deemed useful, use the logarithmic chart provided at the end of the Exam Paper.

[14 Marks]

c) As in Question 3-a), set $d(t) = 0, \forall t$ and $n(t) = 0, \forall t$. Moreover, set $C(s) = 1$ and $T = 1$sec.

Sketch the Nyquist diagram and establish whether the Nyquist closed-loop stability criterion can be used to analyse the closed-loop stability properties.

Hint: to answer Questions 3-c), answering Question 3-a) and 3-b) is not required.

[8 Marks]

